EXP 10: Expectation-Maximization (EM) Algorithm

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
X = np.array([
   [1, 2], [1.5, 1.8], [2, 2.5],
   [7, 8], [8, 8], [8.5, 9]
])
model = GaussianMixture(n_components=2, random_state=0)
model.fit(X)
labels = model.predict(X)
print("Cluster labels:", labels)
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis',
        s=100, edgecolor='black')
plt.scatter(model.means_[:, 0], model.means_[:, 1],
        color='red', marker='X', s=200, label='Cluster Centers')
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Expectation-Maximization Algorithm")
plt.legend()
plt.grid()
plt.show()
```

OUTPUT

Cluster labels: [1 1 1 0 0 0]

